MILITARY SPECIFICATION

SWITCH, TRANSFER, FREQUENCY STANDARD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, evaluation and test, manufacture, and acceptance requirements for a frequency standard transfer switch, hereinafter referred to as the equipment. The purpose of this equipment is to switch between the outputs of two O-1695/U and the AN/WRR-7, VERDIN receive terminal.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2.1), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-E-15090 | Enamel, Equipment, Light-Gray (Formula No. 111) |
| MIL-E-16400 | Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For |
| MIL-E-17555 | Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of |
| MIL-C-39012 | Connector, Coaxial, Radio Frequency, General Specification For |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |
| MIL-STD-108 | Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment |
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-167-1 | Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited) |
| MIL-STD-454 | Standard General Requirements For Electronic Equipment |
| MIL-STD-461 | Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-81111), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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| | |
|---|---|
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement Of |
| MIL-STD-781 | Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-965 | Parts Control Program |
| MIL-STD-1472 | Human Engineering Design Criteria For Military Systems, Equipment And Facilities |

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

NAVAL SUPPLY SYSTEMS COMMAND

| | |
|---|---|
| NAVSUP-4500 | Consolidated Hazardous Item List |

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The equipment shall conform to the requirements of MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Equipment description.

3.1.2.1 Functional description. The required switching configuration shall be as shown in FIGURE 1. The equipment shall consist of a remote control unit and a relay box unit. The remote control unit shall provide a remote switch to operate the relay. It shall also provide a visual signal to indicate if the primary or standby frequency standard is on-line. Changing the remote control switch from a primary to a standby position shall connect 28 volts direct current (VDC) from a PP-7398/WRR-7 power supply to the relay. This in turn shall connect all time and frequency (T and F) outputs to the standby O-1695/U outputs, taking the primary unit off-line.

3.1.2.2 Hardware description.

3.1.2.2.1 Relay box unit. The relay box unit shall contain a multiple-pole, double-throw relay capable of switching between outputs from primary and standby O-1695/U frequency standards. FIGURE 2 shows a wiring diagram of the relay box. In FIGURE 2, the relay is shown in the normal, deactivated state. Closed relay contacts are designated by a dash, and open contacts are designated by an X. It should be noted that all Type BNC connectors (J2, J3, and J5 through J14), in accordance with MIL-C-39012, shall be isolated from ground, and for these connectors, shields shall be switched as well. All other connectors shall be mounted directly to the metal relay box (ground). Provisions shall be made for switching between all of the T and F outputs shown in FIGURE 1. FIGURE 3 shows a sample enclosure for the relay box unit. Connector mounting and required labeling on the relay box side panels are also shown. FIGURE 4 shows a portion of the

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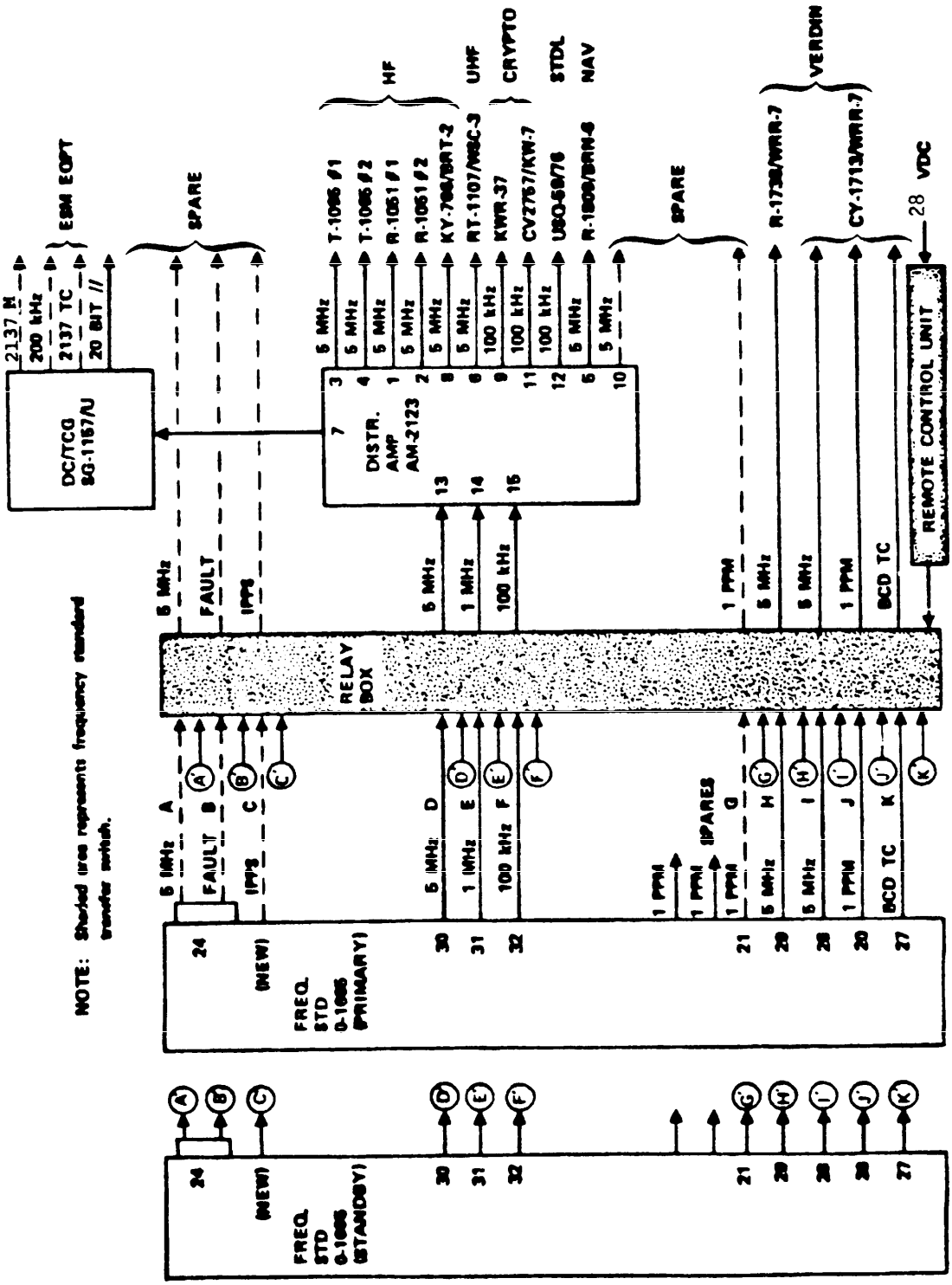



FIGURE 1. T and F distribution system.

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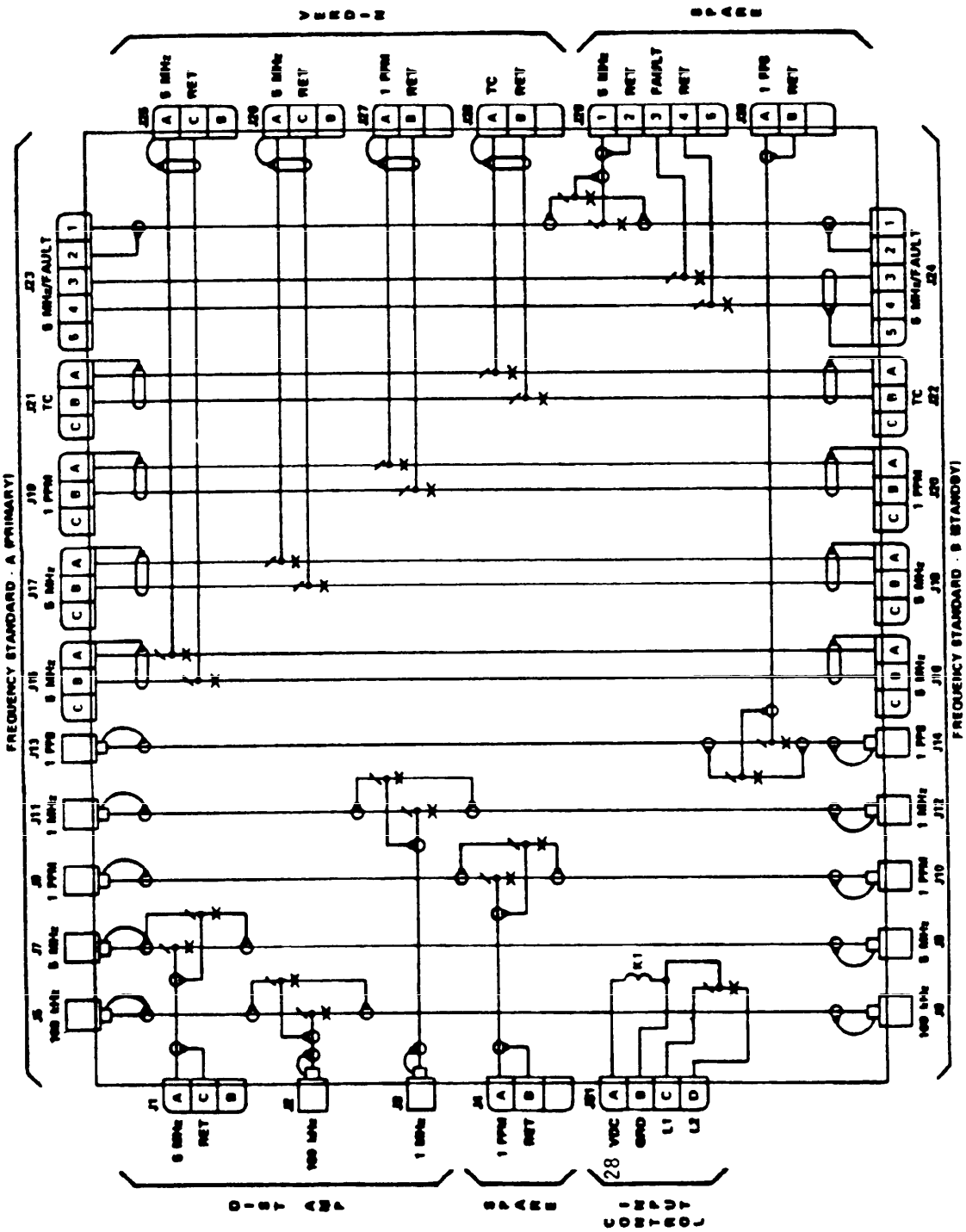



FIGURE 2. Relay box wiring diagram.

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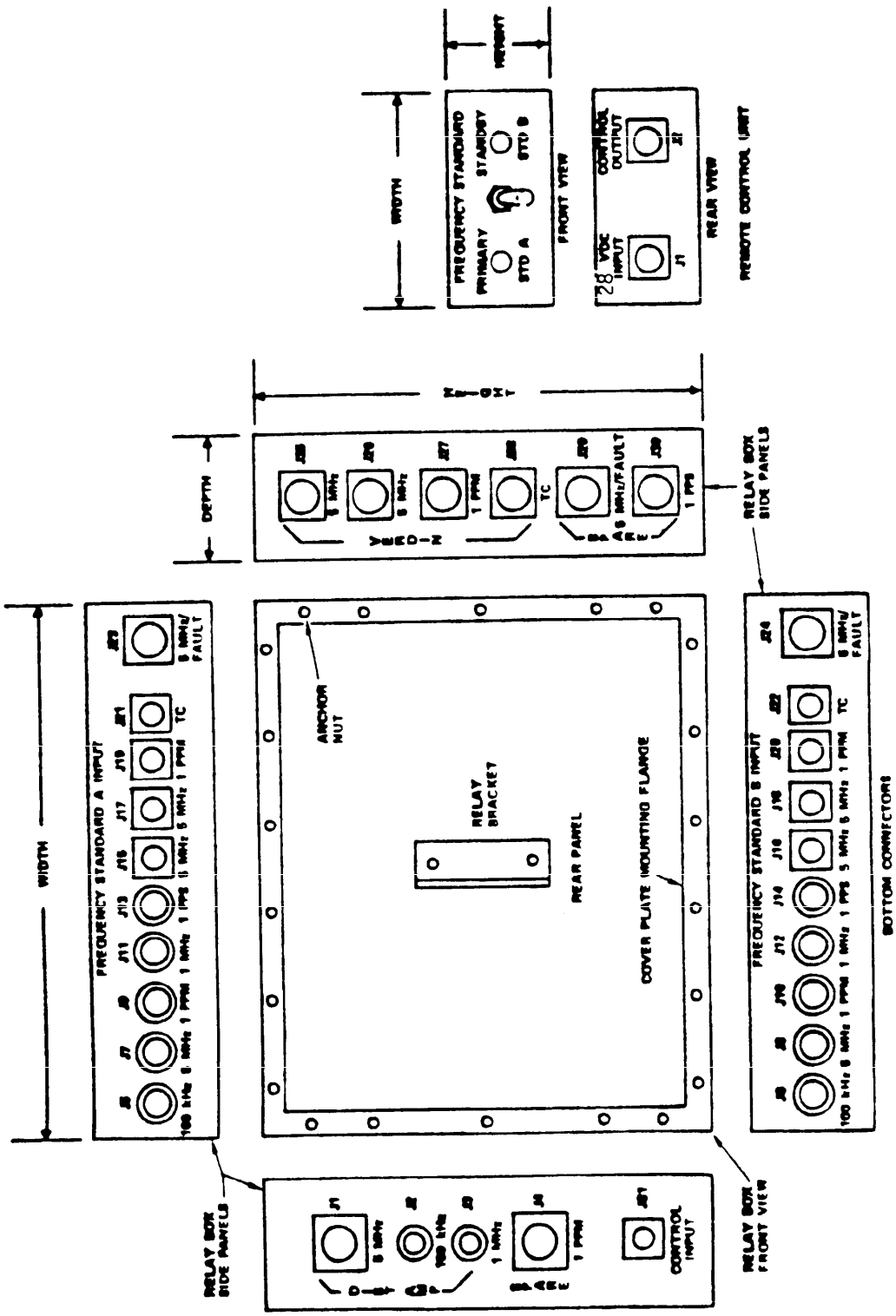



FIGURE 3. Transfer switch enclosures.

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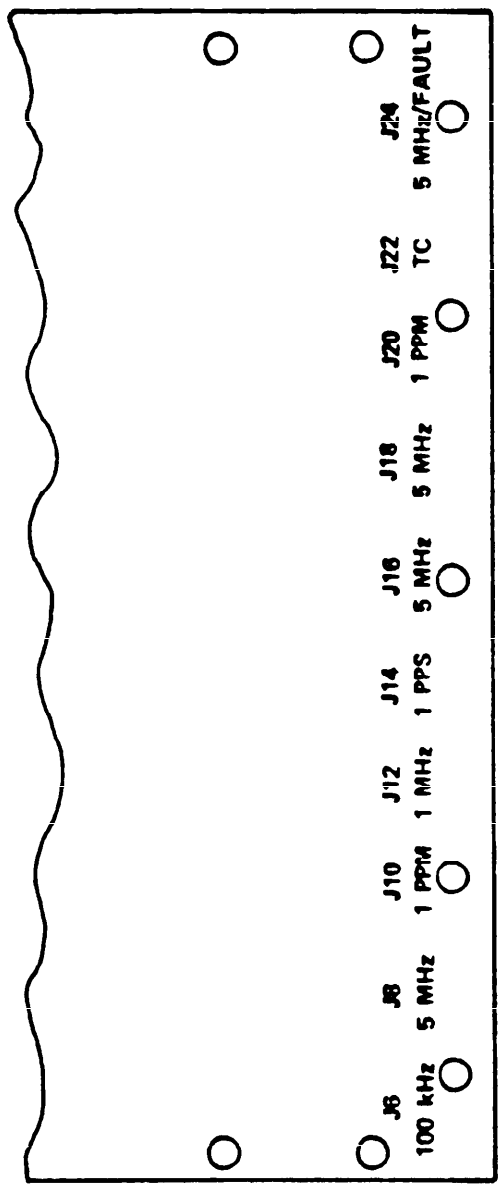



FIGURE 4. Cover plate labeling.

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relay box cover plate showing the labeling to be provided on it for the bottom connectors. The bracket for mounting the relay shall be as shown in FIGURE 3 mounted on the rear panel approximately in the center of the relay box. This relay shall be a 30-pole (minimum), double-throw T-bar, or equivalent, type relay. TABLE I provides relay wiring data for use with a T-bar 931-36CG-28HMS relay.

3.1.2.2.2 Remote control unit. The remote control unit shall be a single-pole, double-throw switch mounted in a small box. The unit shall be equipped with indicator lights to indicate which of the frequency standards is on-line. FIGURE 3 shows a sample enclosure of the unit indicating required connectors and labeling. The K1 relay coil connections shown in FIGURE 2 are interconnected to the remote control unit and 28 VDC power as shown in FIGURE 5. The relay shall be normally deactivated since S1, the MODE switch, is normally open. In this position, 28 VDC from the PP-7398/WRR-7 shall not be connected to the relay coil, but shall be connected through R1 to L1; and L1 shall be grounded through one normally closed contact of the relay. With S1 in the position shown, therefore, the primary standard T and F outputs shall be connected through to equipments as shown in FIGURE 2, and the PRIMARY light, L1, shall be enabled. Closing S1 connects 28 VDC to the relay coil, energizing the relay and connecting all T and F outputs of the standby standard to external equipments. In this case, the primary standard shall be taken off-line. Also, the ground connection to L1 (FIGURE 5) shall be disconnected, turning off the PRIMARY light, and the normally open relay contact shall close, thereby grounding L2 and turning on the STANDBY light. Since the two lights are controlled through relay contacts, they provide visual indication that the relay is operating. Two ten-foot cables, with connectors, shall be supplied as part of the remote control unit for use in connecting this unit to the relay box unit and to the 28 VDC source.

3.1.2.2.3 Interconnections. FIGURE 6, sheets 1 through 3, show the interconnections required for the selector switch. Sheet 1 shows the overall wiring diagram, and sheets 2 and 3 specify detailed interconnections and grounding for each T and F output. FIGURE 6 also specifies the interconnecting cable types that shall be used. Connector types shall be as specified in TABLE II.

3.2 Characteristics.

3.2.1 Performance characteristics.

3.2.1.1 Frequency range. The equipment shall be capable of switching input and output (I/O) signals whose frequencies are within the range of direct current (DC) to 10 megahertz (MHz).

3.2.1.2 Characteristic impedance. The relay box shall be designed to operate within the limits of this specification with I/O connectors terminated in 50 ohms.

3.2.1.3 Crosstalk. The crosstalk between any two connectors shall be less than or equal to 45 decibels (dB) when terminated in 50 ohms and measured at 5 MHz.

3.2.1.4 Isolation. Isolation between adjacent relay contacts shall be at least 35 dB at 5 MHz.

3.2.1.5 Contact power rating. The equipment shall be capable of withstanding continuous input signals at a minimum peak level of 15 volts (V) when terminated in 50 ohms over the frequency range of DC to 10 MHz.

3.2.1.6 Insertion loss. The insertion loss between any two active connectors shall be less than or equal to 0.1 dB at 5 MHz.

3.2.1.7 Intermodulation. The third harmonic of a 5 MHz fundamental tone shall be at least 60 dB below the fundamental.

3.2.1.8 Power source. Maximum power required shall not exceed 10 watts (W). This power will normally be derived from the VERDIN power supply PP-7398/WRR-7. The relay shall activate with control voltages from 15 to 32 VDC.

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TABLE I. Relay wiring connections.

| Junction box | T-bar connector | | T-bar |
|---|---|---|---|
| | Deck | Row | Relay contact |
| J-5 | A | 1 and 2 | Normally closed |
| J-6 | A | 1 and 2 | Normally open |
| J-2 | A | 1 and 2 | Center |
| J-11 | A | 3 and 4 | Normally closed |
| J-12 | A | 3 and 4 | Normally open |
| J-3 | A | 3 and 4 | Center |
| J-7 | A | 5 and 6 | Normally closed |
| J-8 | A | 5 and 6 | Normally open |
| J-1 | A | 5 and 6 | Center |
| J-9 | B | 1 and 2 | Normally closed |
| J-10 | B | 1 and 2 | Normally open |
| J-4 | B | 1 and 2 | Center |
| J-13 | B | 3 and 4 | Normally closed |
| J-14 | B | 3 and 4 | Normally open |
| J-30 | B | 3 and 4 | Center |
| J-15 A White | B | 5 | Normally closed |
| J-15 B Black | B | 6 | Normally closed |
| J-25 A White | B | 5 | Center |
| J-25 C Black | B | 6 | Center |
| J-16 A White | B | 5 | Normally open |
| J-16 B Black | B | 6 | Normally open |
| J-17 A White | C | 1 | Normally closed |
| J-17 B Black | C | 2 | Normally closed |
| J-18 A White | C | 1 | Normally open |
| J-18 B Black | C | 2 | Normally open |
| J-26 A White | C | 1 | Center |
| J-26 C Black | C | 2 | Center |
| J-19 A White | C | 3 | Normally closed |
| J-19 B Black | C | 4 | Normally closed |
| J-20 A White | C | 3 | Normally open |
| J-20 B Black | C | 4 | Normally open |
| J-27 A White | C | 3 | Center |
| J-27 B Black | C | 4 | Center |
| J-21 A White | C | 5 | Normally closed |
| J-21 B Black | C | 6 | Normally closed |
| J-22 A White | C | 5 | Normally open |
| J-22 B Black | C | 6 | Normally open |
| J-28 A White | C | 5 | Center |
| J-28 B Black | C | 6 | Center |

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TABLE I. Relay wiring connections. (Continued)

| Junction box | T-bar connector | | T-bar |
|---|---|---|---|
| | Deck | Row | Relay contact |
| J-23-1 | A | 7 | Normally closed |
| J-23-2 | A | 8 | Normally closed |
| J-23-3 | A | 9 | Normally closed |
| J-23-4 | A | 10 | Normally closed |
| J-29-1 | A | 7 | Center |
| J-29-2 | A | 8 | Center |
| J-29-3 | A | 9 | Center |
| J-29-4 | A | 10 | Center |
| J-24-1 | A | 7 | Normally open |
| J-24-2 | A | 8 | Normally open |
| J-24-3 | A | 9 | Normally open |
| J-24-4 | A | 10 | Normally open |
| J-31 A | - | - | Coil term No. 1 |
| J-31 B | - | - | Coil term No. 2 |
| J-31 B | A | 11 | Center |
| J-31 C | A | 11 | Normally closed |
| J-31 D | A | 11 | Normally open |

Note: Relay connections required to switch shields which do not connect to jack pins are not shown.

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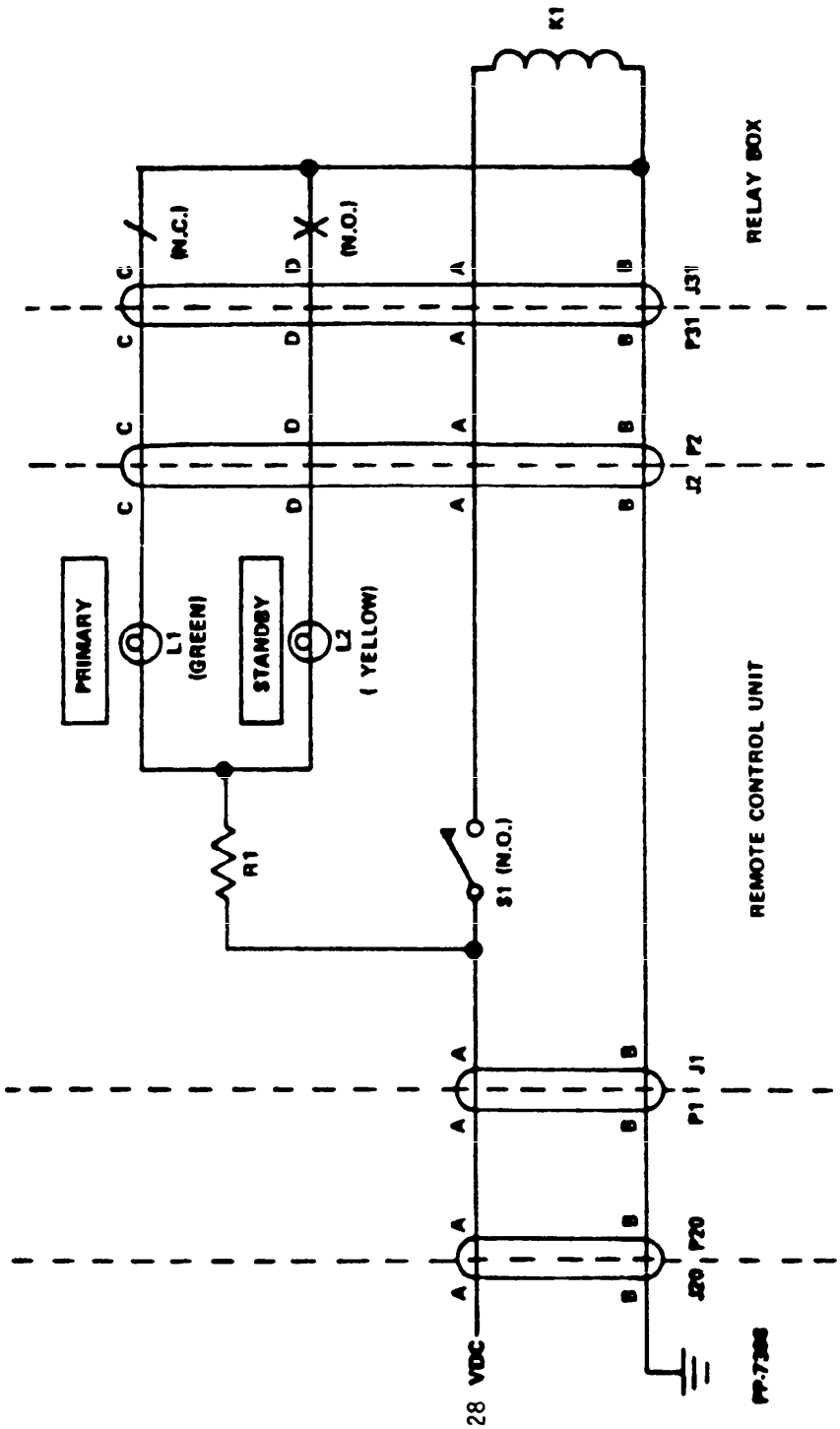
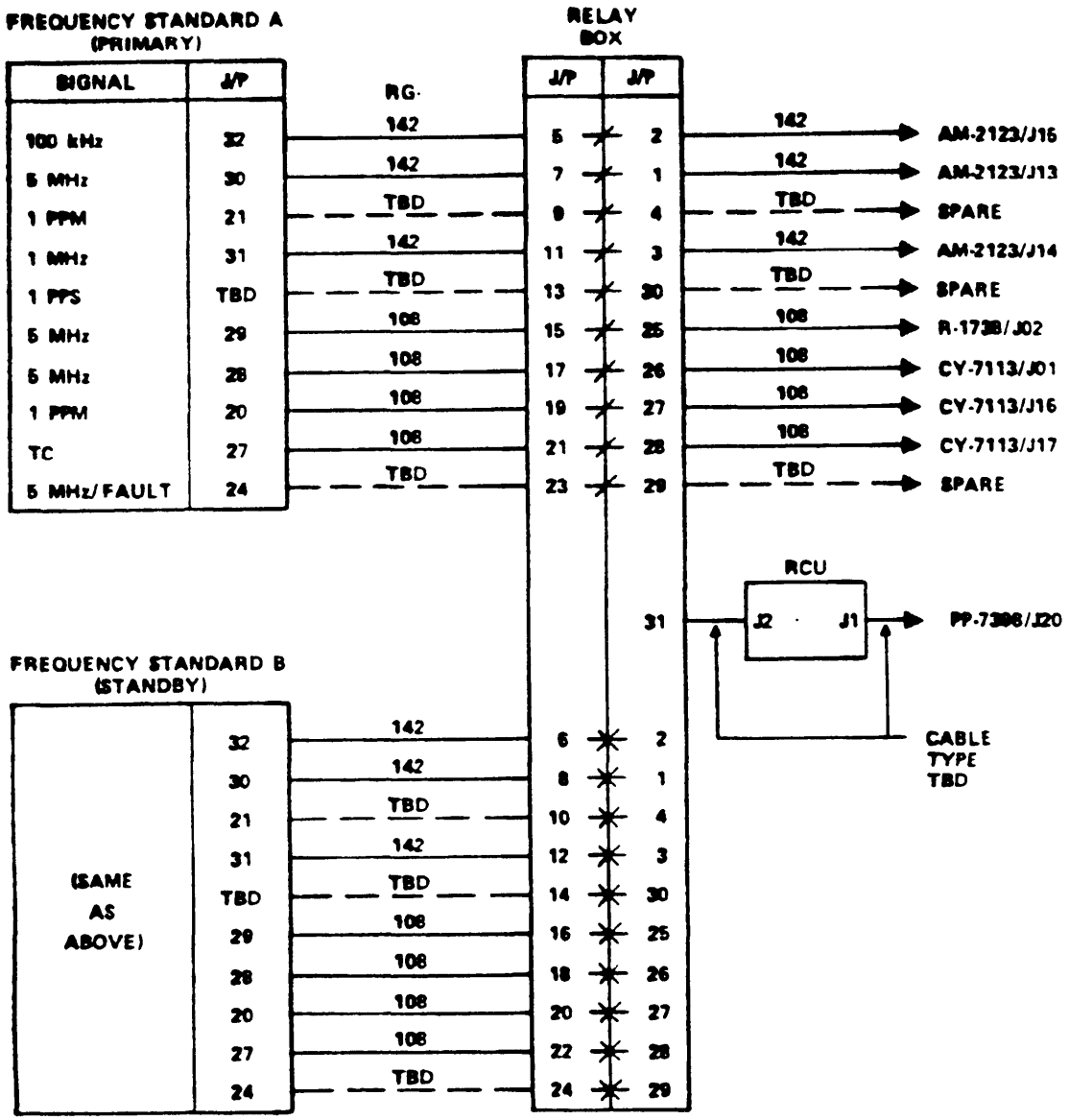



FIGURE 5. Remote control unit wiring diagram.

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NOTES:

1. 1 PPS OUTPUT ON REAR NOT PRESENTLY AVAILABLE.
2. / - NORMALLY CLOSED CONTACT;
   X - NORMALLY OPEN CONTACT.

FIGURE 6. Transfer switch interconnection wiring diagram (sheet 1).

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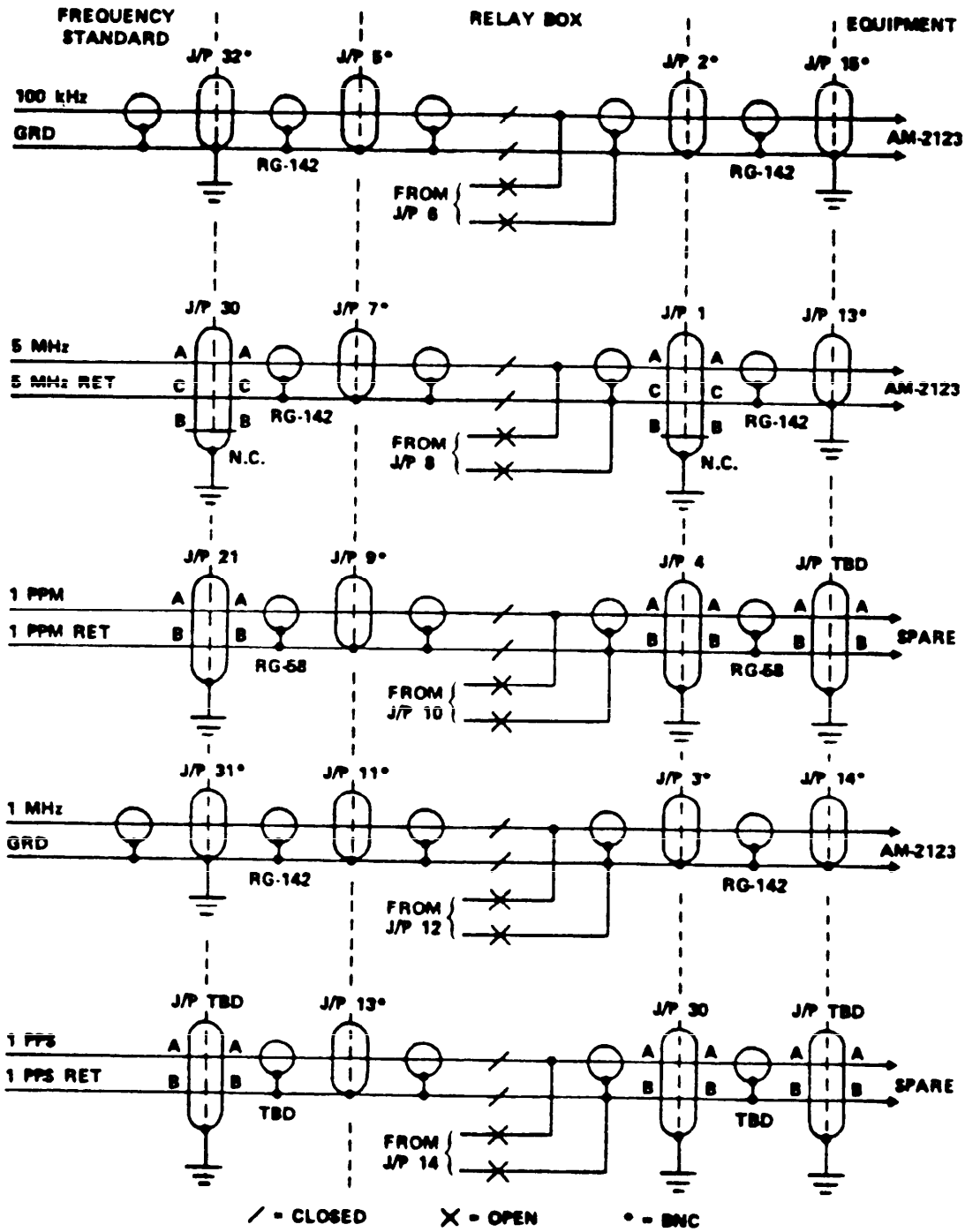



FIGURE 6. Transfer switch interconnection wiring diagram (sheet 2). - Continued

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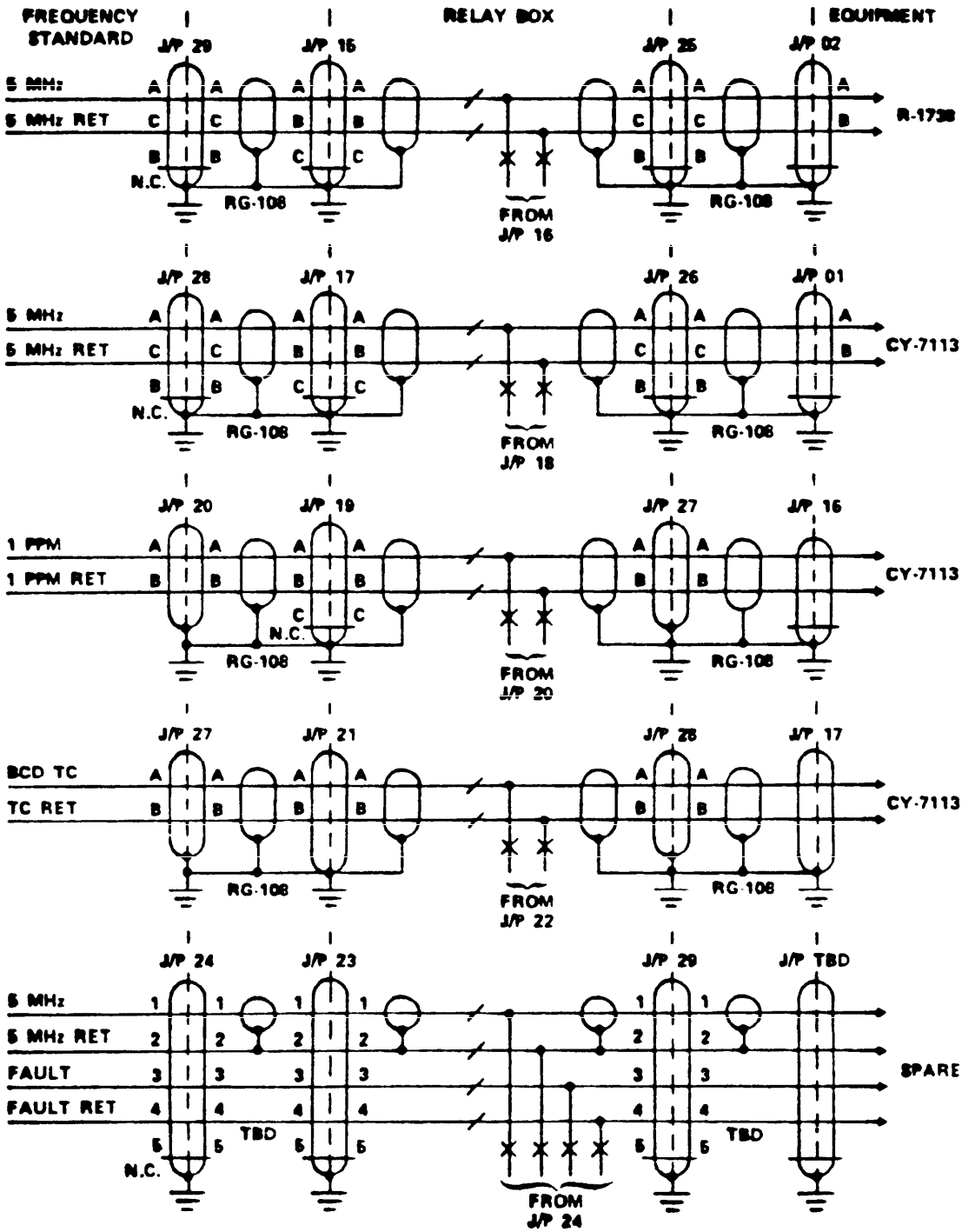



FIGURE 6. Transfer switch interconnection wiring diagram (sheet 3).--Continued

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TABLE II. Connector types.

| FREQUENCY STANDARD | | | | RELAY BOX | | | | |
|---|---|---|---|---|---|---|---|---|
| J | CONN. | P | CONN. | J | CONN. | P | CONN. | REMARKS |
| 32 | AMPH. 31-357-1022 | 32 | AMP 2-392082-2 | 2 | AMP 2-330063-1 | 2 | AMP-2-392082-2 | BNC |
| 30 | MS3452W10SL-3P | 30 | MS3456W10SL-3S & M85049/52-1-10-A | 1 | MS3452W10SL-3P | 1 | MS3456W10SL-3S & M85049/5Z-1-10-A | 3 PINS |
| 21 | MS3452W10SL-4P | | | 4 | MS3452W10SL-4P | | | |
| 31 | AMPH. 31-357-1022 | 21 | MS3456W10SL-4S & M85049/5Z-1-10-A | 3 | AMP 2-330063-1 | 4 | MS3456W10SL-4S & M85049/5Z-1-10-A | 2 PINS |
| TBD | MS3452W10SL-4P | | | 30 | MS3452W10SL-4P | | | |
| 29 | MS3452W10SL-3P | 31 | AMP 2-392082-2 | 25 | MS3452W10SL-3P | 3 | AMP 2-392082-2 | BNC |
| 28 | MS3452W10SL-3P | TBD | MS3456W10SL-4S & M85049/5Z-1-10-A | 26 | MS3452W10SL-3P | 30 | MS3456W10SL-4S & M85049/5Z-1-10-A | 2 PINS |
| 20 | MS3452W10SL-4P | | | 27 | MS3452W10SL-4P | | | |
| 27 | MS3452W10SL-4P | 29 | MS3456W10SL-3S & M85049/52-1-10-A | 28 | MS3452W10SL-4P | 25 | MS3456W10SL-3S & M85049/5Z-1-10-A | 3 PINS |
| 24 | M81511/21EB02S1 | | | 29 | M81511/2EB0S1 | | | |
| | | 28 | MS3456W10SL-3S & M85049/52-1-10-A | 31 | MS3472W8-4P | 26 | MS3456W10SL-3S & M85049/5Z-1-10-A | 3 PINS |
| | | 20 | MS3456W10SL-4S & M85049/5Z-1-10-A | 5,6 | AMP 2-330063-1 | 27 | MS3456W10SL-4S & M85049/5Z-1-10-A | 2 PINS |
| | | 27 | MS3456W10SL-4S & M85049/5Z-1-10-A | 7,8 | AMP 2-330063-1 | 28 | MS3456W10SL-4S & M85049/5Z-1-10-A | 2 PINS |
| | | 24 | M81511/26EB02P1 | 9,10 | AMP 2-330063-1 | 29 | M81511/26EB02P1 | 5 PINS |
| | | | | 11,12 | AMP 2-330063-1 | 31 | MS3476W8-4S & M85049/5Z-1-8-A | |
| | | | | 13,14 | AMP 2-330063-1 | | | |
| | | | | 15,16 | MS3472W8-3P | 5,6 | AMP 2-392082-2 | BNC |
| | | | | 17,18 | MS3472W8-3P | 7,8 | AMP 2-392082-2 | BNC |
| | | | | 19,20 | MS3472W8-3PW | 9,10 | AMP 2-392082-2 | BNC |
| | | | | 21,22 | MS3472W8-3PX | 11,12 | AMP 2-392082-2 | BNC |
| | | | | 23,24 | M81511/2EB0S1 | 13,14 | AMP 2-392082-2 | BNC |
| | | | | | | 15,16 | MS3476W8-3S & M85049/5Z-1-8-A | 3 PINS |
| | | | | | | 17,18 | MS3476W8-3S & M85049/5Z-1-8-A | 3 PINS |
| | | | | | | 19,20 | MS3476W8-3S & M85049/5Z-1-8-A | 3 PINS |
| | | | | | | 21,22 | M3476W8-3SX & M85049/5Z-1-8-A | 3 PINS |
| | | | | | | 23,24 | M81511/26EB02P1 | 5 PINS |

| REMOTE CONTROL UNIT | | | | PP-7398/WRR-7 | | | | |
|---|---|---|---|---|---|---|---|---|
| J | CONN. | P | CONN. | J | CONN. | P | CONN. | |
| 1 | MS3472W8-3P | 1 | MS3476W8-3S & M85049/5Z-1-8-A | 20 | MS3452W10SL-3P | 20 | MS3456VV10SL-3S & M85049/5Z-1-10-A | |
| 2 | MS3472W8-4P | 2 | MS3476W8-4S & M85049/5Z-1-8-A | | | | | |





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3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The maximum dimensions shall not exceed the dimensions specified in TABLE III.

TABLE III. Maximum dimensions.

| Item | Width | | Height | | Depth | | Weight | |
|---|---|---|---|---|---|---|---|---|
| | Centimeters (cm) | Inches (in) | Cm | In | Cm | In | Pounds (lbs) | Kilogram (kg) |
| Relay box | 27.94 | 11 | 22.86 | 9 | 7.62 | 3 | 10 | 4.5 |
| Remote control | 11.43 | 4.5 | 7.62 | 3 | 6.35 | 2.5 | 2 | 0.91 |

3.2.2.2 Connector location and type. Connectors shall be located at the rear of the remote control box and on the side panels of the relay box as shown in FIGURE 3. Types of connectors shall be as specified in TABLE II. Spacing of connectors shall be in accordance with MIL-STD-1472.

3.2.2.3 Connector pin assignments. Connector pin wiring shall be as shown in FIGURES 2 and 5.

3.2.2.4 Maintenance configuration. The relay used in the relay box shall be interconnected in such a way as to ensure that the relay replacement shall not exceed 15 minutes. A suitable connector shall be used for ease of relay replacement.

3.2.2.5 Mounting. The relay box rear panel shall be smooth, without protrusions, and strong enough so that it can be securely fastened to equipment racks by means of user provided mounting holes.

3.2.3 Reliability.

3.2.3.1 Quantitative reliability requirements. The upper test mean-time-between-failures (MTBF) ($\theta_0$ as defined by MIL-STD-781) shall be not less than 10,000 hours. The lower test MTBF ($\theta_1$ as defined by MIL-STD-781) shall be not less than 5000 hours.

3.2.4 Maintainability. The mean-corrective-maintenance-time ($M_{ct}$) shall not exceed one hour.

3.2.5 Environmental conditions. The equipment shall operate within tolerances throughout the specified test without alignment or adjustment. Adherences to the environmental requirements of MIL-E-16400 shall be limited to the requirements specified in 3.2.5.1 through 3.2.5.6.

3.2.5.1 Temperature. The temperature range of the equipment shall be as specified for Range 4 of MIL-E-16400.

3.2.5.2 Humidity. The equipment shall maintain the specified performance when exposed to the humidity conditions of MIL-E-16400.

3.2.5.3 Salt fog (spray). The equipment shall be capable of withstanding the effects of the salt fog test for sheltered equipment as specified in MIL-STD-810, Method 509.1.

3.2.5.4 Vibration. The equipment shall conform to the vibration requirements of MIL-E-16400 and the Type I vibration requirements of MIL-STD-167-1 at vibrational frequencies up to 33 hertz (Hz).

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3.2.5.5 Inclination. The equipment shall conform to the inclination requirements of MIL-E-16400 for submarine installation.

3.2.5.6 Enclosure. The equipment shall conform to the dripproof requirements of MIL-STD-108 when inclined to 45 degrees.

3.2.5.7 Thermal. The equipment shall conform to the thermal requirements of MIL-E-16400.

3.2.5.8 Front panel. The equipment front panel shall be constructed to permit easy access to the rear of the components mounted on the panel and to components located within the enclosure. The design of controls and indicators, and their arrangement on operator and maintenance panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400.

3.3 Design and construction.

3.3.1 Parts, materials, and processes.

3.3.1.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.1.2 Connectors. All connectors and plugs shall be of the type specified in TABLE II. Mating connector plugs shall be provided for all connector receptacles installed on the equipment.

3.3.1.3 Switches. All switches utilized shall conform to Requirement 58 of MIL-STD-454.

3.3.1.4 Indicator lights. All indicators shall conform to Requirement 50 of MIL-STD-454, and shall be visible to a minimum of 100 cm (6 feet (ft)).

3.3.1.5 Wire, hookup, internal. The selection and application of internal electrical hookup wire shall conform to Requirement 20 of MIL-STD-454.

3.3.1.6 Flammable material. Material used as part of the equipment shall conform to Requirement 3 of MIL-STD-454.

3.3.1.7 Painting. The surfaces of the equipment shall conform to the requirements of MIL-E-16400 and MIL-E-15090.

3.3.1.8 Unacceptable materials. Materials specified as unacceptable in MIL-E-16400, those materials defined as hazardous in NAVSUP 4500, and the following listed materials shall not be used in the construction of the equipment:

a. Fiber asbestos electrical insulation
b. Magnesium, magnesium alloy, cast iron
c. Mercury shall not be used in components, processes, tests, or inspections, or come in contact with any part of the equipment at any time.
d. Cadmium plate (electrodeposited)

3.3.2 Electromagnetic interference (EMI) characteristics. The equipment shall conform to the requirements of MIL-STD-461 for Class A5 equipment in respect to RE02, electric field radiated emissions, when carrying working level signals.

3.3.3 Interchangeability. The switches, relays, and connectors of the equipment shall be replaceable and interchangeable, and shall conform to Requirement 7 of MIL-STD-454.

3.3.4 Safety. The equipment shall conform to the Safety criteria paragraph of MIL-E-16400.

3.3.5 Human engineering. The criteria of MIL-STD-1472 shall be used as guidance in the design of the equipments.

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3.3.6 Marking. The equipment shall be marked as shown in FIGURES 4 and 5 and shall conform to Requirement 67 of MIL-STD-454. There shall be no adhesive back labels used in this equipment.

3.3.7 Workmanship.

3.3.7.1 General workmanship. Workmanship shall conform to Requirement 9 of MIL-STD-454. Soldering shall conform to Requirement 5 of MIL-STD-454.

3.3.7.2 Workmanship screen (random vibration). The equipment shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ $\pm 3$ dB from 80 Hz to 350 Hz. Requirement 9 of MIL-STD-454 shall govern the definition and meaning of defect.

3.3.7.3 Workmanship screen (temperature cycling). The equipment shall withstand a temperature cycling screen (see 4.6.5.2) to ensure elimination of initial failures due to random defective components and marginal operation due to component lot variations or poor assembly processes and procedures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection for preparation for delivery (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV.

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4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be made on every equipment offered for delivery. This inspection shall comprise all necessary examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which will detect hidden defects of materials. Production inspection shall include the examinations and tests specified in Group A of TABLE IV.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. This inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests specified in Group B of TABLE IV.

4.4.2.1 Sampling plan for Group B inspection. Equipment submitted for Group B inspection shall be selected by the Government quality assurance representative (QAR) and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptance quality level (AQL) shall be 6.5 percent defective and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor shall rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

TABLE IV. Examinations and tests.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Surface examination | | 4.6.2 | X | X | | |
| Size | 3.2.2.1 | 4.6.2 | X | X | | |
| Parts, materials, and processes | 3.3.1 | 4.6.2 | X | X | | |
| Finish | 3.3.1.7 | 4.6.2 | X | X | | |
| Marking | 3.3.6 | 4.6.2 | X | X | | |
| Safety | 3.3.4 | 4.6.2 | X | X | | |
| Workmanship | 3.3.7 | 4.6.2, 4.6.4 | X | X | | |
| Workmanship screen | 3.3.7.1, 3.3.7.2 | 4.6.5.1, 4.6.5.2 | X<br>X | X<br>X | | |
| Temperature | | | | | | |
| Low temperature | 3.2.5.1 | 4.6.8.1.1 | X | | | X |
| High temperature | 3.2.5.1 | 4.6.8.1.2 | X | | | X |
| Humidity | 3.2.5.2 | 4.6.8.2 | X | | | X |
| Salt fog | 3.2.5.3 | 4.6.8.3 | X | | | X |
| Vibration | 3.2.5.4 | 4.6.8.4 | X | | | X |
| Inclination | 3.2.5.5 | 4.6.8.5 | X | | | X |

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TABLE IV. Examinations and tests. (Continued)

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Enclosure | 3.2.5.6 | 4.6.9 | X | | X | |
| Performance tests | 3.2.1 | 4.6.3 | X | X | | |
| Crosstalk | 3.2.1.3 | 4.6.10.1 | X | X | | |
| Isolation | 3.2.1.4 | 4.6.10.2 | X | X | | |
| Contact power rating | 3.2.1.5 | 4.6.10.3 | X | X | | |
| Insertion loss | 3.2.1.6 | 4.6.10.4 | X | X | | |
| Intermodulation | 3.2.1.7 | 4.6.10.5 | X | X | | |
| Power Source | 3.2.1.8 | 4.6.10.6 | X | X | | |
| EMI | 3.3.2 | 4.6.11 | X | | | X |
| Reliability | 3.2.3 | 4.7 | X | | | |
| Maintainability | 3.2.4 | 4.8 | X | | | |
| Connectivity | 3.1.2.2 | 4.6.12 | X | | X | |
| Interchangeability | 3.3.3 | 4.6.6 | X | | | X |
| Thermal test | 3.2.5.7 | 4.6.7 | X | | | X |

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, such as extremes of temperature and humidity, effect of salt air; and tests of the effects of vibration and inclination. Environmental inspection shall include the examinations and tests specified in Group C of TABLE IV.

4.4.3.1 Sampling for environmental inspection. One equipment shall be randomly selected from each successive production lot of 50 equipments, or less, as required by the Government. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection in 4.4.3, the contractor shall immediately investigate the cause of failure and shall implement the corrective action necessary to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The equipment shall then be retested.

4.4.4 Reinspection of conforming production control and environmental test units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to and have passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection specified in 4.4.1 after repair of all defects and damages.

4.5 Test equipment and test conditions. Test equipment and test conditions for first article inspection and quality conformance inspection shall conform to the following quality assurance provisions of MIL-E-16400:

a. Test facilities, chambers, equipment, and apparatus
b. Standard test conditions
c. Performance of test

4.6 Test methods.

4.6.1 Examinations and tests. The examinations and tests specified in TABLE IV shall be performed in accordance with the applicable test methods specified in 4.6.2 through 4.6.12.

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4.6.2 Surface examination. The equipment shall be examined to ensure compliance with the following requirements:

a. Size (see 3.2.2.1)
b. Parts, materials, and processes (see 3.3.1)
c. Finish (see 3.3.1.7)
d. Marking (see 3.3.6)
e. Safety (see 3.3.4)
f. Workmanship (see 3.3.7)

4.6.3 Performance test. To the extent specified in TABLE IV, the equipment shall be subjected to a performance test including all operating controls to determine compliance with the performance parameters and safety requirements specified herein. The performance tests shall be as specified herein and in 4.6.10.

4.6.4 General workmanship examination. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.3.4.1.

4.6.5 Workmanship screen.

4.6.5.1 Vibration. Prior to conducting temperature cycling, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random, or subject to contracting activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 7. The control accelerometer shall be located next to one of the mounting points of the item under vibration. Equipment having no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

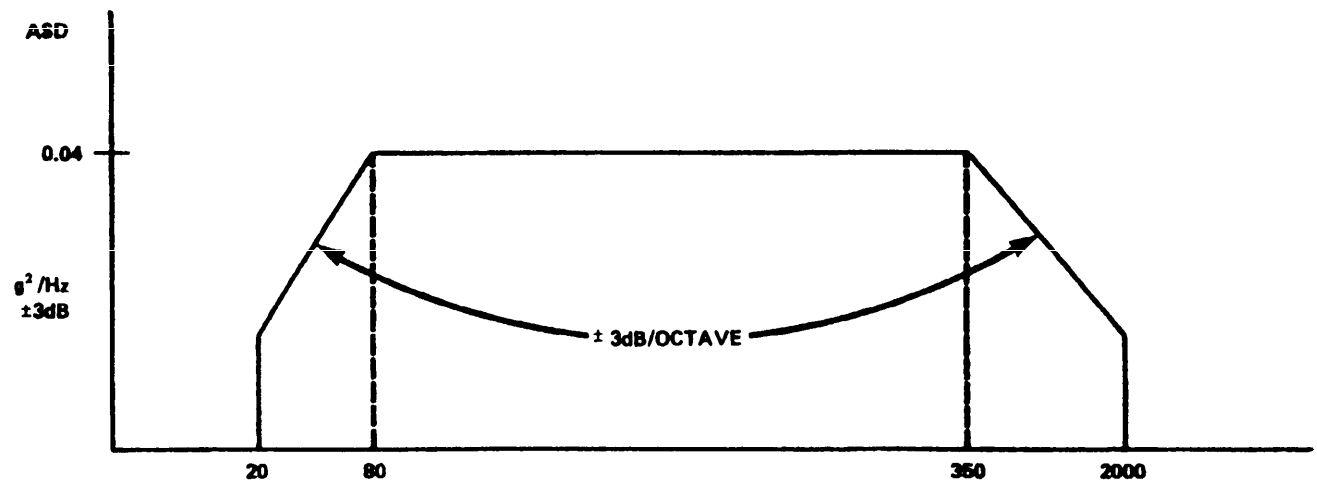



FIGURE 7. Random vibration curve.

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4.6.5.2 Temperature cycling. Each equipment shall be subjected to 10 cycles of the temperature curve shown in FIGURE 8. The temperature rate of change shall be not less than 5° Celsius (C) per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit reaches 80 percent of the chamber temperature. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. When failures occur, the equipment shall be reworked and the cycling continued for a cumulative total of 10 cycles.

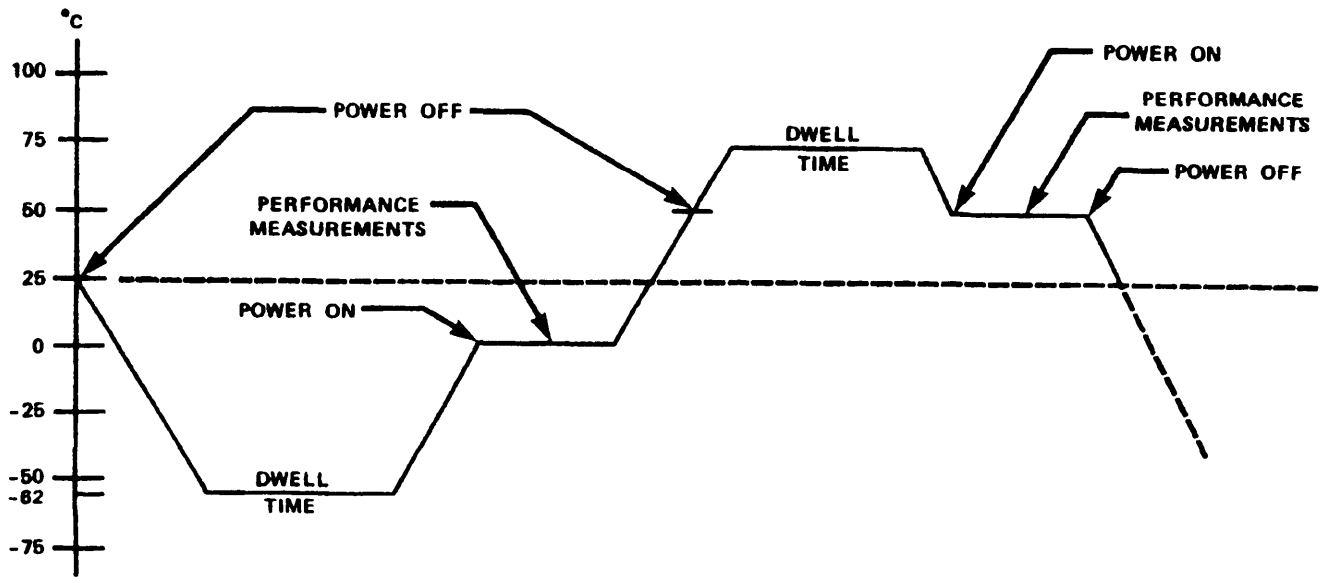



FIGURE 8. Temperature curve (one cycle).

4.6.6 Interchangeability. The requirements of 3.3.5 shall be demonstrated on two switchboards randomly chosen from each lot of 50 or less. Satisfactory equipment performance shall be demonstrated prior to, and after, the interchange of the interchangeable and replaceable switches, relays, and connectors.

4.6.7 Thermal survey test. A thermal survey shall be performed on the equipment in accordance with MIL-E-16400.

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4.6.8 Environmental tests.

4.6.8.1 Temperature test sequence. The operating and nonoperating test sequence shall be as specified in 4.6.8.1.1 and 4.6.8.1.2.

4.6.8.1.1 Low temperature. The low temperature test shall be performed in accordance with Method 502.1 of MIL-STD-810, except as follows:

a. Step 2. Storage or nonoperating temperature shall be -62°C $^{+0°C}_{-5°C}$ for a duration of 24 hours.
b. Step 4. Temperature Range 4: 0°C to +50°C
c. The combined duration of Steps 4 and 5 shall be not less than 12 hours, at least two of which shall be in the stabilized condition prior to conducting tests.

4.6.8.1.2 High temperature. The high temperature test shall be performed in accordance with Method 501.1, Procedure I, of MIL-STD-810, except as follows:

a. Step 4. Applicable high operating temperature for the range specified in 3.2.5.1.
b. The combined duration of Steps 4 and 5 shall be not less than 12 hours, at least six of which shall be in the stabilized condition prior to conducting tests.
c. Step 7. The equipment shall operate for a period of at least two hours in the stabilized condition before Step 7 tests are performed.

4.6.8.2 Humidity. Humidity tests shall be performed in accordance with Method 507.1, Procedure IV, of MIL-STD-810, except as follows:

a. Step 5. Equipment shall undergo the humidity cycle test shown in FIGURE 507.1-3 of MIL-STD-810 except the high temperature limit shall be 50°C.
b. Step 9. The equipment shall be carefully examined in detail to detect evidence of failure. Blistering of organic coatings shall be a cause for rejection.

4.6.8.3 Salt fog sheltered equipment. The salt fog test specified in Method 509.1 of MIL-STD-810 shall be applied to the finishes and coatings on parts and frame and enclosure structures as finally assembled for use. Sample corner structures and any other critical sections may be used for the test. The test shall not be applied to the complete equipment.

4.6.8.4 Vibration. A vibration test shall be performed in accordance with MIL-E-16400, except that the vibrational frequency range shall be 4 to 33 Hz.

4.6.8.5 Inclination. An inclination test shall be performed in accordance with MIL-E-16400 for submarines.

4.6.9 Enclosure tests. Enclosure tests shall be performed in accordance with MIL-STD-108. For dripproof enclosures, the actual drip test shall be conducted. For first article inspection, the enclosure test shall be conducted after completion of the vibration, inclination, and shock tests.

4.6.10 Electrical tests. The equipment shall be subjected to the tests specified in 4.6.10.1 through 4.6.10.6.

4.6.10.1 Crosstalk. FIGURE 9 shows the test set-up to be used to measure crosstalk. All other 5 MHz lines except the ones shown in FIGURE 9 shall be disconnected. Also, the configuration shown permits measurement on an adjacent relay contact. This, along with the fact that the highest frequency of interest is used, gives a worst-case crosstalk measurement. Voltage $E_1$ shall be measured first, and then $E_2$ shall be measured with J25 terminated in 50 ohms. Crosstalk measures relative to $E_1 - E_2$ shall be less than 45 dB.

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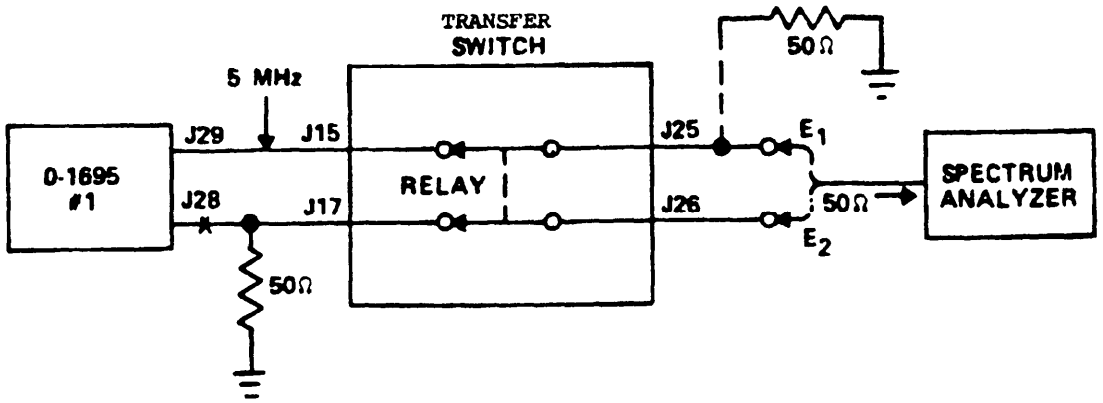



FIGURE 9. Crosstalk test set-up.

4.6.10.2 Isolation. FIGURE 10 shows the test set-up to be used to measure isolation. It should be noted that, with this set-up, isolation shall be measured across adjacent relay contacts, giving worst-case values. Also, all 5 MHz lines except the one being measured shall be disconnected. The procedure shall be to measure $E_1$, the reference signal, and then to activate the relay and measure $E_2$. The radio frequency (RF) signal line isolation values shall be computed in dB as $E_1 - E_2$. For 5 MHz, isolation shall be less than 35 dB.

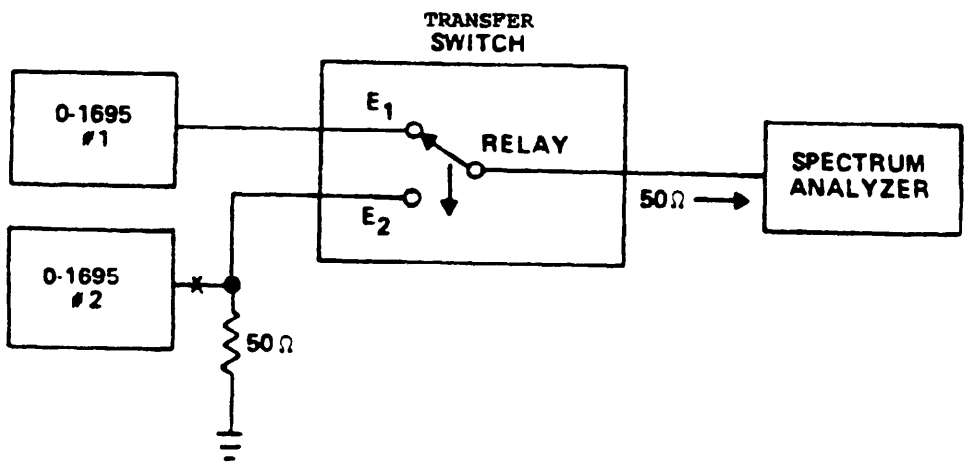



FIGURE 10. Isolation test set-up.

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4.6.10.3 Contact test. The contact test shall be performed prior to the insertion loss tests. With each output terminal terminated into 50 ohms and with appropriate voltages on input terminals, the relay shall be switched off and on 100 times per test. This test shall be performed twice, first with greater than 15 VDC on each input terminal and then a second time with a greater than 10.6 V root-mean-square (rms) 10 MHz signal on each input. At the end of each test, the contacts shall be examined. Pitted or any mechanically degraded contacts shall be cause for rejection.

4.6.10.4 Insertion loss. FIGURE 11 shows the test set-up to be used for measuring insertion loss. Insertion loss shall be less than 0.1 dB. Insertion loss test shall be performed after the contact test.

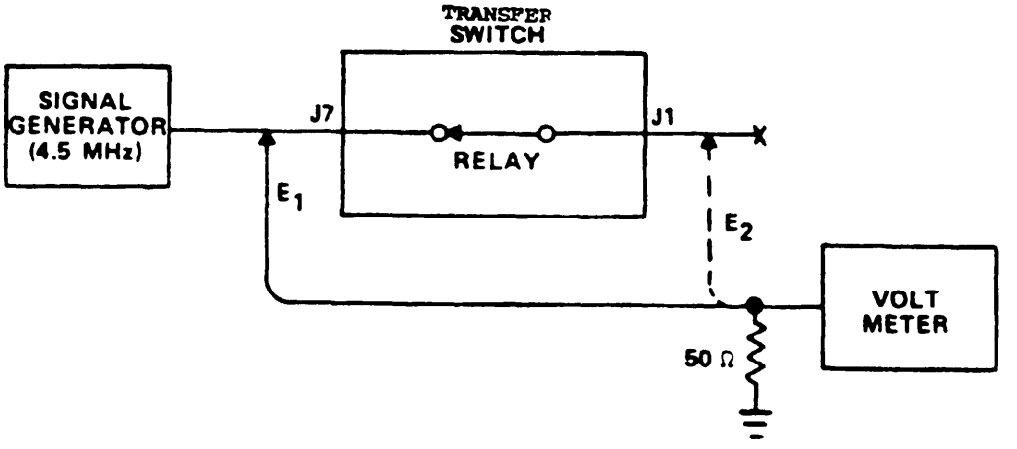



FIGURE 11. Insertion loss test set-up.

4.6.10.5 Intermodulation test. With a 5 MHz fundamental frequency on the input, the third harmonic shall be a minimum of 60 dB below the fundamental frequency. This test shall be performed for each I/O combination with the relay both in the ON and OFF state.

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4.6.10.6 ower rating test. While using a 28 V, ±1 V, power source and with the relay switch activated, the power lead's current shall be measured. The power lead shall draw no greater than 300 milliamperes (mA). The relay shall activate with control voltages from 15 to 32 VDC.

4.6.11 Electromagnetic radiation tests. The requirements of 3.3.2 shall be verified by tests performed in accordance with MIL-STD-462. For the frequency standard selection switch, only test RE02, radiated emissions, electric field, shall be performed. This test shall be performed at the frequencies generated by the cesium beam frequency reference, 0-1695/U.

4.6.12 Connectivity. Continuity checks shall be made between all relay box I/O connectors, with the remote control unit switch in each position, to determine proper switching operation as shown in FIGURE 2.

4.7 Reliability demonstration. A reliability demonstration shall be performed in accordance with the following:

a. Test temperature shall be maintained at 50°C.
b. Vibrational stress (continuous) shall be as specified in the Combined environments for shipboard equipment paragraph of MIL-STD-781.
c. The electrical load shall consist of:
   1. Frequency range of DC to 5 MHz inclusive
   2. Impedance 50 ohms nominal
   3. Activating voltage 24 to 32 VDC
d. The relays shall be switched on and off 120 times with 15 VDC on each input terminal.
e. The switching rate shall continue for a period of five days (120 hours). If no failures occur during this time the equipment will have satisfactorily passed the reliability test.
f. The switching rate shall be once every 30 seconds.

4.8 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment $M_{ct}$ requirements of 3.2.4 are fulfilled.

4.9 Inspection of preparation for delivery. Inspections shall be conducted to ensure compliance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use in submarines.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Use of other than current documents listed in the DoDISS (see 2.1)

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c. Number of first article samples to be submitted if other than specified in 4.3
d. When reinspected production control and environmental sample units may not be accepted (see 4.4.4).
e. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified herein shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of the Defense Acquisition Regulation (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified herein shall be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

| Paragraph | Data requirement | Applicable DID |
|---|---|---|
| 4.2 | Procedures, Test | UDI-T-22710 |
| 4.3, 4.4 | Test Report | DI-T-5247 |
| 4.3, 4.4 | Report, Failure Item Analysis | DI-R-7039 |
| 4.6.7 | Report, Thermal Survey | DI-R-7036 |
| 4.7 | Reports, Reliability Test and Demonstration | DI-R-7034 |
| 4.7 | Procedures, Reliability Tests | DI-R-7035 |
| 4.8 | Plan, Maintainability Demonstration | DI-R-2129 |
| 4.8 | Report, Maintainability Demonstration | DI-R-2130 |
| 4.6.11 | Report, Test, Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC) | DI-R-2066/R021 |
| 4.4 | Report, Failure Summary and Analysis | DI-R-7041 |
| 4.6.2 e | Report, Safety Assessment | DI-R-7049 |

(Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

Preparing activity:
NAVY-EC
(Project 6625-N795)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-28867(EC) Switch, Transfer, Frequency Standard

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO. |
|---|---|
| | DATE |

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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